

Intern Learning & Progress Tracker — Software Requirements Specification

Project type: Internal web application **Version:** 1.0 (baseline for Sprint 1 kickoff) **Duration:** 6–8 weeks, three 2-week sprints (plus Sprint 0 setup week) **Tracking:** PearlTrack **Team:** 5 interns — MERN (1), Java (1), Data Science (2), QA (1) — with a mentor acting as Product Owner **Status:** Draft for Product Owner sign-off

1. Document Purpose and How to Use It

This document expands the high-level project brief into a buildable backlog. Every requirement has:

- a **stable ID** (use it in PearlTrack ticket titles, commit messages, and test cases),
- an **owner domain** (MERN / JAVA / DS / QA / ALL),
- a **priority**: **M** = MVP (Sprints 1–2), **S3** = Sprint 3, **ST** = Stretch,
- **acceptance criteria** that QA converts directly into test cases.

ID prefixes:

Prefix	Area
FR-AUTH	Authentication, sessions, RBAC
FR-USR	User, cohort, and mentor-mapping management
FR-CRS	Course catalog and enrollment
FR-ASG	Assignments and submissions

FR-MLS	Milestones
FR-PRG	Intern progress logging
FR-FBK	Mentor review and feedback
FR-DSH	Dashboards (intern / mentor / L&D)
FR-NTF	Notification service (Java)
FR-RPT	Reporting service (Java)
FR-SCR	Progress-scoring model (DS-1)
FR-INS	Insights dashboard (DS-2)
FR-ADM	Admin and configuration
FR-SYS	Cross-cutting platform behaviour
NFR	Non-functional requirements
QA	Quality and test requirements
INT	Integration contracts between domains

Assumption note: where the brief was silent, this document proposes a concrete decision and marks it [ASSUMPTION]. The Product Owner should confirm or change each one in Section 16 before Sprint 1 planning. Nothing here is meant to be accepted silently.

2. Glossary

Term	Meaning
Intern	A learner in the programme. Logs progress, submits assignments, receives feedback.
Mentor	Reviews assigned interns' work, gives feedback, receives digests. One mentor may have many interns.

L&D / Admin	Learning & Development staff. Full read access across cohorts, manages catalog and users, consumes analytics.
Cohort	A named intake of interns with a start and end date (e.g. "Summer 2026 Batch A").
Course	A named learning unit in the catalog (internal or external, e.g. a Udemy course).
Enrollment	The assignment of a Course to an Intern, with a due date and completion state.
Assignment	A deliverable an intern must submit (task, exercise, capstone). May or may not belong to a course.
Submission	An intern's response to an assignment; may have multiple attempts.
Milestone	A dated checkpoint grouping courses/assignments (e.g. "Week 4 — Fundamentals complete").
Progress Log	An intern's self-reported entry of effort and completion against a course or assignment.
Risk Band	Model output: on-track, watch, at-risk.
Digest	A scheduled summary email sent to a mentor or L&D.
Skill Tag	A capability label attached to courses and assignments, used for skill-gap analysis.

3. Stakeholders and Personas

3.1 Intern ("Priya", first-week joiner)

Needs to know what is assigned to her, what is due next, and whether she is behind. Logs 5–15 minutes of activity per day. Will abandon the tool if logging takes more than ~30 seconds per entry. **Design implication:** one-screen dashboard, quick-log widget, no mandatory

fields beyond the minimum.

3.2 Mentor ("Arun", reviews 4–6 interns)

Has limited time. Wants a single queue of things waiting on him, sorted by urgency, and wants to give feedback without navigating three screens. **Design implication:** a review queue with inline feedback; weekly digest email that is actionable, not decorative.

3.3 L&D / Admin ("Meera")

Cares about the cohort, not the individual — until someone is at risk. Wants exportable evidence for programme reviews. **Design implication:** cohort overview first, drill-down second; exports on every table.

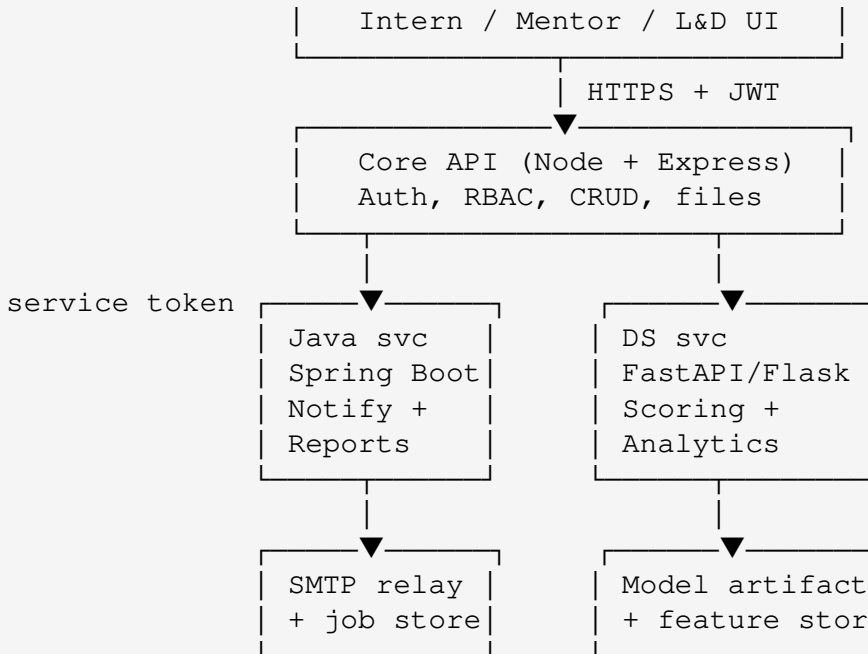
3.4 Secondary stakeholders

- **Product Owner (mentor)** — prioritises the backlog, accepts stories.
- **Scrum Master (rotating intern)** — runs ceremonies, maintains PearlTrack hygiene.
- **Future intern batches** — the tool must be re-usable, so cohort data must be isolated and the catalog must be reusable across cohorts.

4. System Architecture

4.1 Component overview

React SPA (MERN)



MongoDB (primary store, owned exclusively by C

4.2 Architectural rules (non-negotiable — these are the integration contract)

ID	Rule
ARCH-1	MongoDB is owned by the Core API only. The Java and DS services never connect to the database directly. They read through Core API endpoints. This keeps the schema free to change and makes each service independently testable.
ARCH-2	All inter-service calls use a service account JWT signed with a shared secret, carrying the claim <code>role: service</code> and a <code>svc</code> name. No service uses an end-user token.
ARCH-3	Every service exposes <code>GET /health</code> (liveness) and <code>GET /health/ready</code> (dependencies checked).

ARCH-4	All APIs are versioned under <code>/api/v1/</code> . Breaking changes require a new version path.
ARCH-5	All timestamps are stored and transmitted in UTC ISO-8601 . The UI converts to the browser's local timezone.
ARCH-6	Contracts are agreed before implementation and published as an OpenAPI 3 spec per service, committed to the repo. Consumers build against mocks generated from the spec.
ARCH-7	The MERN app must degrade gracefully when Java or DS is unavailable: the page renders, the affected panel shows an inline "temporarily unavailable" state, and no request blocks page load.

4.3 Technology stack **[ASSUMPTION]**

Layer	Choice	Notes
Frontend	React 18 + Vite, React Router, TanStack Query, Tailwind CSS	Component library optional; keep dependencies minimal
Core API	Node 20 + Express, Mongoose	
Database	MongoDB 6	Single replica for dev; Atlas free tier acceptable
Java service	Java 17 + Spring Boot 3, Spring Scheduler (or Quartz), Thymeleaf for email templates	
Email	SMTP relay (internal) or Mailhog in dev	No third-party email SaaS without approval
DS service	Python 3.11 + FastAPI, pandas, scikit-learn	

File storage	Local volume in dev, S3-compatible bucket in shared env	
CI	GitHub Actions (or equivalent)	Runs lint, unit tests, Newman, Playwright smoke
Containers	Docker + docker-compose for the full stack	One command must bring up all four services

4.4 Repository strategy [ASSUMPTION]

Single monorepo with four top-level folders: `/web`, `/core-api`, `/notification-service`, `/analytics-service`, plus `/qa` (test assets) and `/docs` (specs, ADRs, this document). Rationale: five interns, one demo day, and cross-service contract changes that need to land atomically.

Branching: trunk-based with short-lived feature branches named `<domain>/<ticket-id>-<slug>`, e.g. `mern/ILT-142-mentor-review-queue`. PR requires one reviewer and green CI.

5. Roles and Permissions

5.1 Role definitions

Role	Description
INTERN	Owens their own data only.
MENTOR	Read/write over interns explicitly mapped to them.
ADMIN	L&D. Full access within the organisation, including user management and catalog.

SERVICE

Machine role for Java/DS services. Read-only on aggregate endpoints plus write access to its own domain (notifications, scores).

5.2 Permission matrix

○ = own records only · A = assigned interns only · ✓ = all records · — = no access

Capability	INTERN	MENTOR	ADMIN	SERVICE
View own dashboard	○	○	○	—
View intern profile	○	A	✓	✓
Create/edit/delete course in catalog	—	—	✓	—
View course catalog	✓	✓	✓	✓
Enroll an intern in a course	—	A	✓	—
Un-enroll / change due date	—	A	✓	—
Create/edit assignment	—	A	✓	—
Submit assignment	○	—	—	—
View submission	○	A	✓	✓

Give feedback on submission	—	A	✓	—
Log progress entry	O	—	—	—
Edit/delete own progress entry (within 24h)	O	—	✓	—
Create milestone	—	A	✓	—
View mentor review queue	—	O	✓	—
View cohort overview	—	A (own interns only)	✓	✓
Manage users / roles	—	—	✓	—
Map intern → mentor	—	—	✓	—
Trigger report generation	—	A	✓	✓
View analytics dashboard	O (own scores only)	A	✓	—
View risk score and reasons	O (band only, no raw score) [ASSUMPTION]	A (full)	✓	✓
Configure notification rules	—	O (own prefs)	✓	—
View audit log	—	—	✓	—

FR-AUTH-14 note: whether interns see their own risk band is a genuine policy decision, not a technical one. Default is "yes, band only, with a supportive message and no numeric score" — the PO must confirm. Showing an intern a raw "0.23 at-risk" number without context is more likely to demoralise than help.

6. Functional Requirements

6.1 Authentication, Session, and Access Control — MERN

ID	Pri	Requirement	Acceptance criteria
FR-AUTH-01	M	Email + password login	Valid credentials return access token (15 min) + refresh token (7 days, httpOnly cookie) and the user's role and profile. Invalid credentials return HTTP 401 with a generic message that does not reveal whether the email exists.
FR-AUTH-02	M	Password storage	bcrypt, cost factor ≥ 12. Plaintext or reversible storage is a release blocker.
FR-AUTH-03	M	Password policy	Min 10 chars, at least one letter and one number, blocked against a common-password list. Enforced server-side, mirrored client-side.
FR-AUTH-04	M	Logout	Refresh token revoked server-side; subsequent refresh returns 401.
FR-AUTH-05	M	Token refresh	POST /auth/refresh issues a new access token if the refresh token is valid and unrevoked. Rotates the refresh token.

FR-AUTH-06	M	Role-based route guarding (server)	Every endpoint declares required role(s). A request with insufficient role returns 403 and is written to the audit log. Server-side enforcement is authoritative; UI hiding is cosmetic.
FR-AUTH-07	M	Role-based UI rendering	Navigation, buttons, and routes not permitted for the user's role are not rendered.
FR-AUTH-08	M	Ownership checks	Mentors can only act on interns in their mapping; interns only on their own records. Enforced in a shared middleware, not per-controller.
FR-AUTH-09	M	First-login forced password change	Admin-created accounts start with a temporary password and <code>mustChangePassword: true</code> ; all routes except the change-password route redirect until changed.
FR-AUTH-10	S3	Password reset by email	Single-use token, 30-minute expiry, invalidated on use or on a newer request.
FR-AUTH-11	M	Brute-force protection	5 failed attempts in 15 minutes locks the account for 15 minutes; lockout logged and visible to admin.
FR-AUTH-12	M	Session timeout	Idle session (no API call) for 60 minutes forces re-login. A warning modal appears at 55 minutes.
FR-AUTH-13	ST	SSO / Google Workspace login	Only if the organisation's IdP is available to the team; otherwise out of scope.
FR-AUTH-14	S3	Risk visibility policy toggle	Admin setting controls whether interns see their own risk band. Default per Section 5.2.

6.2 User, Cohort, and Mentor Mapping — MERN

ID	Pri	Requirement	Acceptance criteria
FR-USR-01	M	Admin creates users	Fields: full name, email (unique), role, cohort, mentor (if intern), start date, status. Duplicate email returns 409.
FR-USR-02	M	Admin edits and deactivates users	Deactivation is soft — the user cannot log in, but their historical data, submissions, and feedback remain intact and still appear in cohort history. Hard delete is not available.
FR-USR-03	M	Bulk user import via CSV	Template downloadable. Import is transactional per row: valid rows import, invalid rows are reported with row number and reason in a downloadable error file.
FR-USR-04	M	Intern → mentor mapping	One mentor per intern (primary). Reassignment preserves historical feedback attribution to the original mentor.
FR-USR-05	S3	Secondary / backup mentor	Optional second mentor with review rights but not included in digest ownership counts.
FR-USR-06	M	Cohort CRUD	Name, start date, end date, status (planned / active / completed), L&D owner. A cohort with active interns cannot be deleted, only closed.
FR-USR-07	M	Profile page	All roles can view and edit their own name, avatar, timezone, and notification preferences. Email and role are admin-only.

FR-USR-08	M	User list with search and filter	Filter by role, cohort, mentor, status. Server-side pagination, 25 per page.
	S3	Mentor workload view	Admin sees interns-per-mentor count and pending-review count, to rebalance assignments.
FR-USR-10	ST	Cohort cloning	Copy a completed cohort's course/assignment plan into a new cohort without copying people or progress. This is what makes the tool reusable for the next batch.

6.3 Course Catalog and Enrollment — MERN

ID	Pri	Requirement	Acceptance criteria
FR-CRS-01	M	Course CRUD (admin)	Fields: title, description, provider, external URL, skill tags (≥ 1), estimated hours, difficulty (beginner/intermediate/advanced), category, active flag.
FR-CRS-02	M	Skill tag management	Admin maintains a controlled tag vocabulary. Free-text tags are not allowed — skill-gap analysis (FR-INS-03) is worthless against inconsistent tags.
FR-CRS-03	M	Course catalog browse	All roles can browse, search by title, and filter by tag/difficulty/category.
FR-CRS-04	M	Enroll intern(s) in a course	Mentor or admin selects one or more interns and a due date. Duplicate active enrollment for the same intern+course is rejected with a clear message.

FR-CRS-05	M	Enrollment states	not_started → in_progress → completed, plus dropped. Transitions are logged with actor and timestamp.
FR-CRS-06	M	Completion percentage	Intern sets 0–100 in steps of 5 via progress logging. Reaching 100 sets status completed and stamps completedAt.
FR-CRS-07	M	Due date and overdue flag	An enrollment is overdue when now > dueDate and status ≠ completed. The flag is derived, never stored stale.
FR-CRS-08	S3	Bulk enrollment by cohort	Assign a course to an entire cohort in one action, with a shared due date, skipping anyone already enrolled.
FR-CRS-09	S3	Course prerequisites	Optional list of prerequisite courses; the UI warns (does not block) when enrolling an intern who has not completed them.
FR-CRS-10	M	Learning path / course ordering	Courses within a cohort plan carry a sequence number to drive the intern's "what's next" panel.
FR-CRS-11	S3	Certificate/ evidence upload on completion	Intern attaches a certificate file or URL when marking a course complete; mentor sees it during review.

6.4 Assignments and Submissions — MERN

ID	Pri	Requirement	Acceptance criteria
----	-----	-------------	---------------------

FR-ASG-01	M	Assignment CRUD	Fields: title, description (rich text), optional parent course, type (<code>task/exercise/project/quiz</code>), due date, max score, skill tags, attachments, assigned-to (individuals or whole cohort).
FR-ASG-02	M	Assignment visibility	An intern sees only assignments targeted at them or their cohort, and only after the assignment's publish date.
FR-ASG-03	M	Draft vs published	Assignments are created as drafts; interns see nothing until published. Editing a published assignment's due date notifies affected interns (see FR-NTF-06).
FR-ASG-04	M	Submission	Intern submits with: notes, optional file attachment(s), optional external link (repo/doc URL). At least one of notes/file/link is required.
FR-ASG-05	M	Late submission	Allowed by default and flagged <code>isLate: true</code> with the delay in hours recorded. Admin can configure an assignment to hard-close at the due date.
FR-ASG-06	M	Resubmission	Permitted after a mentor requests changes. Each attempt is retained with an incrementing <code>attemptNo</code> ; history is viewable.
FR-ASG-07	M	Submission states	<code>not_submitted</code> → <code>submitted</code> → <code>(under_review)</code> → <code>changes_requested</code> → <code>resubmitted</code> → <code>approved</code> .

FR-ASG-08	M	File upload constraints	Max 10 MB per file, 3 files per submission. Allowed: pdf, doc/docx, ppt/pptx, png, jpg, zip, txt, md. MIME type validated server-side, not just by extension. Files stored outside the web root with non-guessable names.
FR-ASG-09	M	File download authorisation	A file is only downloadable by its owner, their mentor, or an admin. Direct URL access by anyone else returns 403. This is a specific, testable security case — QA should attempt it explicitly.
FR-ASG-10	S3	Assignment templates	Reusable assignment definitions that can be instantiated into a new cohort.
FR-ASG-11	M	My assignments list	Intern view filtered by status, sorted by due date ascending, with overdue items pinned to the top and visually distinct.

6.5 Milestones — MERN

ID	Pri	Requirement	Acceptance criteria
FR-MLS-01	M	Milestone CRUD	Title, description, target date, scope (cohort-wide or individual), linked courses and assignments.
FR-MLS-02	M	Auto-derived milestone status	completed when all linked items are complete; at_risk when the target date is within 3 days and completion < 60%; missed when the date has passed with items outstanding; otherwise on_track.

FR-MLS-03	M	Milestone timeline widget	Intern dashboard shows the next three milestones with a progress bar each.
FR-MLS-04	S3	Mentor sign-off on milestones	A milestone may require explicit mentor confirmation before counting as complete.

6.6 Progress Logging — MERN

ID	Pri	Requirement	Acceptance criteria
FR-PRG-01	M	Quick log entry	Intern logs: date (defaults to today), linked course or assignment, hours spent (0.5 increments), new completion %, notes, optional blocker flag with description. Submitting takes ≤ 3 interactions from the dashboard.
FR-PRG-02	M	Validation	Hours between 0.5 and 12 per entry; date not in the future; completion % cannot decrease without a reason note.
FR-PRG-03	M	Edit window	An intern may edit or delete their own entry within 24 hours; after that only an admin can correct it, and the correction is audit-logged. This protects the integrity of the DS training data.
FR-PRG-04	M	Progress history	Chronological list, filterable by course/date range, with a weekly hours total.
FR-PRG-05	M	Blocker flag	Setting a blocker raises the item in the mentor's review queue and triggers a notification (FR-NTF-05).

FR-PRG-06	S3	Streak / consistency indicator	Days-logged-this-week counter on the intern dashboard. Framed positively (encouragement), never punitively.
FR-PRG-07	M	Derived fields for analytics	Each entry stores <code>loggedAt</code> (server time) separately from <code>date</code> (user-declared), so submission-delay features are computable.

6.7 Mentor Review and Feedback — MERN

ID	Pri	Requirement	Acceptance criteria
FR-FBK-01	M	Review queue	Single screen listing everything awaiting the mentor: new submissions, blockers, overdue items, milestone sign-offs. Sortable by due date, intern, and age of request. Default sort: oldest pending first.
FR-FBK-02	M	Inline review	Mentor opens a submission, sees the intern's notes/files/links plus their recent progress logs in the same view, and submits feedback without leaving the page.
FR-FBK-03	M	Feedback content	Rating 1–5 (required), comments (required, min 20 chars), optional action items list, decision (<code>approve / request_changes</code>).

FR-FBK-04	M	Feedback visibility	Visible to the intern immediately on save. Mentors cannot edit feedback after 1 hour; corrections are added as a new feedback entry so the conversation history stays honest.
FR-FBK-05	M	Intern acknowledgement	Intern marks feedback as read; unacknowledged feedback older than 5 days appears in the mentor digest.
FR-FBK-06	S3	Feedback threads	Intern can reply once per feedback item to ask a clarifying question; mentor can respond. Not a full chat — two levels deep maximum.
FR-FBK-07	S3	Feedback templates / canned comments	Mentor-managed snippets to reduce review time.
FR-FBK-08	M	Bulk acknowledge / no-action	Mentor can dismiss a queue item with a one-click "no action needed" that records the review without full feedback.
FR-FBK-09	ST	AI-generated feedback draft	Suggests a first-draft comment from the submission and progress history; the mentor must edit and explicitly approve before it is saved. Clearly labelled as a draft in the UI. Never auto-sent.

6.8 Dashboards — MERN (consuming Java + DS)

Intern dashboard

ID	Pri	Requirement
FR-DSH-01	M	Header summary: courses completed / total, assignments pending, overdue count, hours logged this week.
FR-DSH-02	M	"Due next" panel: the five nearest deadlines across courses, assignments, and milestones, colour-coded by urgency.
FR-DSH-03	M	Recent feedback panel with unread badge.
FR-DSH-04	M	Quick-log widget embedded on the dashboard (FR-PRG-01).
FR-DSH-05	S3	Personal progress chart: cumulative completion vs expected pace over time.
FR-DSH-06	S3	Own risk band with supportive guidance, subject to FR-AUTH-14.

Mentor dashboard

ID	Pri	Requirement
FR-DSH-10	M	Assigned-interns table: name, courses complete %, assignments pending, overdue count, last activity date, risk band.
FR-DSH-11	M	Pending-review count with direct link to the queue.
FR-DSH-12	M	Intern drill-down: full profile with progress history, submissions, feedback given, and milestone status.
FR-DSH-13	S3	At-risk interns highlighted at the top with the model's top contributing factors shown.
FR-DSH-14	S3	Trigger an on-demand digest for a single intern.

L&D / Admin overview

ID	Pri	Requirement
FR-DSH-20	M	Cohort selector, then: headcount, average completion %, overdue assignment count, submissions this week, active mentors.
FR-DSH-21	M	Cohort progress table: every intern, sortable by completion, overdue count, and last activity.
FR-DSH-22	M	Mentor activity table: reviews completed, average turnaround time, pending count. Framed as workload balancing, not individual performance ranking.
FR-DSH-23	S3	Risk distribution summary (on-track / watch / at-risk counts) from the DS service.
FR-DSH-24	S3	Embedded insights dashboard panels (FR-INS-*).
FR-DSH-25	M	CSV export on every table.
FR-DSH-26	ST	PDF export of the cohort overview (via Java reporting service).
FR-DSH-27	M	Empty and loading states for every panel; a failed DS/Java call shows an inline message and never blanks the page (ARCH-7).

6.9 Notification Service — JAVA

ID	Pri	Requirement	Acceptance criteria
----	-----	-------------	---------------------

FR-NTF-01	S3	Weekly mentor digest	Scheduled job, default Monday 08:00 in the mentor's timezone. Contains, per assigned intern: completion delta this week, submissions awaiting review, overdue items, blockers raised, risk band. Sent even if empty? No — skipped with a log entry when there is nothing to report, to avoid digest fatigue.
FR-NTF-02	S3	Overdue assignment flagging	Daily job at 07:00 identifies assignments past due and un-submitted. Notifies the intern on day 1 and day 3 overdue, and the mentor on day 3 and day 7. Escalation to L&D at day 14. No daily nagging.
FR-NTF-03	S3	Upcoming deadline reminder	Notification 48 hours before an assignment or course due date.
FR-NTF-04	S3	New feedback notification	Sent to the intern within 5 minutes of feedback being saved.
FR-NTF-05	S3	Blocker alert	Sent to the mentor within 5 minutes of a blocker being flagged.
FR-NTF-06	S3	Assignment change notification	Sent when a published assignment's due date or description changes materially.
FR-NTF-07	S3	In-app notification centre	Bell icon with unread count; notification list with read/unread state and deep links. Served from the Java service, rendered by MERN.
FR-NTF-08	S3	User notification preferences	Per-user, per-type toggles for email and in-app. Critical account notifications (password reset, lockout) cannot be disabled.

FR-NTF-09	S3	Delivery log and retry	Every notification records status (queued/sent/failed/skipped) with the reason. Failed email sends retry 3 times with exponential backoff, then mark failed and surface in the admin health view.
FR-NTF-10	S3	Idempotency	A digest for a given (user, period) is generated at most once. Re-running the job does not double-send. QA must test this by re-triggering the job manually.
FR-NTF-11	S3	Manual trigger endpoint	Admin-only endpoint to run any scheduled job immediately, for demo day and for testing. Essential — do not make QA wait until Monday 08:00 to test the digest.
FR-NTF-12	S3	Email templates	Responsive HTML with plain-text fallback, organisation branding, and an unsubscribe/preferences link. Rendered from the intern's/mentor's actual data — no placeholder text in production templates.
FR-NTF-13	ST	Slack/Teams webhook channel	Same notification payloads delivered to a channel webhook.

6.10 Reporting Service — JAVA

ID	Pri	Requirement	Acceptance criteria
FR-RPT-01	S3	Cohort progress report	Parameters: cohort, date range. Output: per-intern completion, overdue counts, average feedback rating, hours logged. Formats: JSON (for MERN), CSV.

FR-RPT-02	S3	Individual intern report	Full history: enrollments, submissions, feedback, progress logs, milestones.
FR-RPT-03	S3	Mentor activity report	Reviews completed, average turnaround, pending backlog, per mentor.
FR-RPT-04	S3	Overdue report	All currently overdue items across a cohort with age in days.
FR-RPT-05	ST	PDF rendering	Same reports rendered to PDF with cover page, generation timestamp, and page numbers.
FR-RPT-06	S3	Async generation for large reports	Reports over a size threshold return a job ID; the client polls for completion and then downloads. Prevents request timeouts on demo day.
FR-RPT-07	S3	Report history	Generated reports listed with parameters, requester, and timestamp; downloadable for 30 days.
FR-RPT-08	S3	Scheduled report delivery	Admin can schedule the cohort report to be emailed weekly or monthly.
FR-RPT-09	S3	Caching	Report data cached for 15 minutes to avoid hammering the Core API during a demo. Cache is bypassable with a <code>refresh=true</code> parameter.

6.11 Progress Scoring Model — DS (Data Scientist 1)

ID	Pri	Requirement	Acceptance criteria
----	-----	-------------	---------------------

FR-SCR-01	S3	Feature engineering pipeline	Computes, per intern per scoring date: course completion rate, completion rate vs cohort median, average submission delay (hours), late-submission ratio, average feedback rating, feedback rating trend, days since last activity, logging consistency (days logged / days elapsed), blockers raised in last 14 days, overdue item count, hours logged vs estimated hours.
FR-SCR-02	S3	Rule-based baseline (ship this first)	A transparent weighted-threshold scorer that works from day one, before any model is trained. This is the Sprint 3 deliverable; the ML model is the stretch. Deliver the baseline early so MERN can integrate against a real response, not a mock.
FR-SCR-03	ST	Trained classification model	Logistic regression or gradient-boosted trees producing a probability plus band. Documented train/test split, class balance handling, and hyperparameters.
FR-SCR-04	S3	Risk banding	on-track / watch / at-risk with documented, configurable thresholds. Bands are what users see; the raw score is for mentors and admins.

FR-SCR-05	S3	Explainability	Every score returns the top 3 contributing factors in plain English (e.g. "4 assignments overdue", "no activity logged in 9 days"). A score without a reason is not shippable — a mentor cannot act on a bare number, and an unexplained "at-risk" label attached to a named intern is unfair to that intern.
FR-SCR-06	S3	Scheduled batch scoring	Nightly re-score of all active interns. Results stored with <code>asOfDate</code> and <code>modelVersion</code> so history is queryable.
FR-SCR-07	S3	On-demand scoring endpoint	<code>POST /scoring/interns/{id}/refresh</code> recomputes immediately.
FR-SCR-08	S3	Score history	Trend of an intern's score over time, so mentors can see whether an intervention worked.
FR-SCR-09	ST	Model evaluation report	Accuracy, precision, recall, F1, confusion matrix on a held-out set, plus a note on which errors matter more. False negatives (missing a struggling intern) are worse than false positives here; tune the threshold accordingly and say so in the report.
FR-SCR-10	S3	Synthetic data generator	Generates a realistic multi-cohort dataset (≥ 60 interns $\times$ 8 weeks) for development and model training. Blocking dependency: a live cohort of five interns over six weeks produces nowhere near enough data to train or demo anything. Build this in Sprint 1, not Sprint 3.

FR-SCR-11	S3	Minimum-data guard	If an intern has fewer than 5 activity records or has been active fewer than 7 days, return band <code>insufficient_data</code> rather than a misleading score.
FR-SCR-12	S3	Model/version metadata endpoint	Returns model version, training date, feature list, and thresholds, so the UI can display "scored by v1.2 on 4 Oct".
FR-SCR-13	S3	Fairness and use-policy note	A short written statement in <code>/docs</code> : scores are a prompt for a mentor conversation, never an automated judgement; they are not used for hiring, extension, or performance decisions. Surface a one-line version of this in the UI next to any risk band.

6.12 Insights Dashboard — DS (Data Scientist 2)

ID	Pri	Requirement	Acceptance criteria
FR-INS-01	S3	Cohort trend analysis	Cumulative completion over time, cohort vs expected pace, weekly submission volume, weekly active-logger count.
FR-INS-02	S3	Time-to-completion per course	Median, mean, and spread of days from enrollment to completion, per course; highlights courses that consistently overrun their estimated hours.

FR-INS-03	S3	Skill-gap analysis	Aggregates skill tags across completed vs assigned vs available courses to show which capabilities the cohort has covered and which are thin. Requires the controlled tag vocabulary (FR-CRS-02).
FR-INS-04	S3	Engagement heatmap	Day-of-week × week-of-programme logging activity, to spot drop-off points in the programme design.
FR-INS-05	S3	Feedback score distribution	Distribution of ratings by course and over time. Presented at cohort level; per-mentor comparison is deliberately excluded to avoid the dashboard becoming a mentor-ranking tool.
FR-INS-06	S3	Bottleneck identification	Courses and assignments with the highest overdue or resubmission rates.
FR-INS-07	S3	Filters	Cohort, date range, mentor, skill tag, course — applied consistently across all panels.
FR-INS-08	S3	Delivery format	Chart-data JSON endpoints consumed and rendered by the MERN frontend, so there is one UI and one auth model. [ASSUMPTION — the alternative, a separate Streamlit/Dash app, is faster for DS but creates a second login and a fragmented demo. If the PO prefers speed over integration, flag it in Sprint 1, not Sprint 3.]

FR-INS-09	S3	Aggregation floor	Any aggregate computed over fewer than 5 interns is suppressed and shown as "not enough data", so cohort-level charts cannot be used to infer one identifiable person's performance.
FR-INS-10	ST	Cohort comparison	Compare two cohorts side by side on the same metrics — the payoff for reusing the tool across batches.
FR-INS-11	ST	Scheduled insight snapshot	Weekly snapshot of key metrics persisted so historical trends survive data changes.

6.13 AI-Generated Summaries — ST (shared MERN + DS)

ID	Pri	Requirement
FR-AI-01	ST	Weekly intern summary: a short natural-language paragraph generated from the week's activity, shown on the intern dashboard and included in the mentor digest.
FR-AI-02	ST	Mentor digest narrative: a 3–5 sentence overview across all assigned interns, highlighting who needs attention and why.
FR-AI-03	ST	Every generated summary is labelled as AI-generated in the UI and in emails.
FR-AI-04	ST	Summaries are derived only from data already visible to the recipient — no cross-intern leakage into another intern's summary.
FR-AI-05	ST	A generation failure or timeout degrades to the existing structured digest; it never blocks the email.

6.14 Admin and Configuration — MERN

ID	Pri	Requirement
FR-ADM-01	M	Admin settings screen: organisation name, logo, default timezone, academic week start day.
FR-ADM-02	S3	Notification rule configuration: digest day/time, overdue escalation thresholds, reminder lead time.
FR-ADM-03	S3	Risk threshold configuration: band cut-offs, editable by admin, versioned so historical scores remain interpretable.
FR-ADM-04	M	Audit log viewer: filter by actor, entity type, action, date range.
FR-ADM-05	S3	System health panel: status of Core API, Java service, DS service, email delivery, and last successful job run for each scheduled job. Genuinely useful on demo day.
FR-ADM-06	S3	Data export (admin): full cohort data as CSV/JSON for archival.

6.15 Cross-Cutting Platform Requirements

ID	Pri	Requirement
FR-SYS-01	M	Consistent API error envelope: <code>{ "error": { "code": "VALIDATION_FAILED", "message": "...", "details": [...], "traceId": "..." } }</code> . All three services use the same shape.
FR-SYS-02	M	Server-side validation on every write endpoint, with field-level error details returned to the client.
FR-SYS-03	M	Pagination on every list endpoint: <code>?page=&limit=</code> (default 25, max 100), response includes <code>total</code> , <code>page</code> , <code>totalPages</code> .
FR-SYS-04	M	Audit logging for: login/logout/failed login, role change, user create/deactivate, enrollment change, feedback create, progress-log admin edit, permission denial.

FR-SYS-05	M	Structured JSON application logging with a correlation ID propagated across all three services via an <code>X-Request-Id</code> header.
FR-SYS-06	M	Soft delete for all user-generated content; nothing is physically removed within the programme lifetime.
FR-SYS-07	M	Seed script producing a demo-ready dataset: 1 admin, 4 mentors, 30 interns, 12 courses, 40 assignments, 6 weeks of realistic progress logs and feedback. One command, repeatable, idempotent.
FR-SYS-08	M	Every screen has defined loading, empty, error, and no-permission states. These are requirements, not polish — QA writes test cases against them.
FR-SYS-09	S3	Global search across interns, courses, and assignments, scoped by the user's permissions.

7. Data Model

MongoDB collections. `_id` is an `ObjectId` on every document; `createdAt` / `updatedAt` / `createdBy` / `updatedBy` / `isDeleted` are present on every collection and omitted below for brevity.

users

```

email           string, unique, lowercase, indexed
passwordHash    string
fullName        string
role            enum: INTERN | MENTOR | ADMIN
cohortId        ObjectId → cohorts (interns only)
mentorId        ObjectId → users (interns only)
secondaryMentorId ObjectId → users, nullable
status          enum: active | inactive | pending
timezone        string, IANA, default "Asia/Kolkata"
avatarUrl       string, nullable

```

```
startDate          date
mustChangePassword boolean
lastLoginAt        datetime
failedLoginCount   int
lockedUntil         datetime, nullable
notificationPrefs  object
```

Indexes: {email:1} unique, {role:1,cohortId:1},
{mentorId:1,status:1}

cohorts

```
name, description, startDate, endDate
status      enum: planned | active | completed
ldOwnerId  ObjectId → users
```

courses

```
title, description, provider, externalUrl
skillTags          [ObjectId → skillTags], min 1
estimatedHours     number
difficulty         enum: beginner | intermediate | advanced
category           string
sequenceNo        int, nullable
prerequisites      [ObjectId → courses]
isActive           boolean
```

Index: {skillTags:1}, text index on title+description

skillTags

```
name, unique
category
description
```

enrollments

internId	ObjectId → users, indexed
courseId	ObjectId → courses, indexed
assignedBy	ObjectId → users
assignedAt	datetime
dueDate	date
status	enum: not_started in_progress complete
completionPct	int 0-100, step 5
startedAt	datetime, nullable
completedAt	datetime, nullable
evidenceUrl	string, nullable

Index: {internId:1, courseId:1} unique on active records,
{dueDate:1, status:1}

assignments

title, description (rich text/HTML, sanitised)	
courseId	ObjectId → courses, nullable
cohortId	ObjectId → cohorts, nullable
assignedTo	[ObjectId → users] // empty = whole cohort
type	enum: task exercise project quiz
skillTags	[ObjectId → skillTags]
dueDate	datetime
publishAt	datetime
maxScore	number, nullable
allowLate	boolean, default true
attachments	[fileRef]
status	enum: draft published archived

submissions

assignmentId	ObjectId, indexed
internId	ObjectId, indexed

attemptNo	int
submittedAt	datetime
isLate	boolean
delayHours	number
notes	string
attachments	[fileRef]
externalLink	string, nullable
status	enum: submitted under_review changes_r

Index: {assignmentId:1, internId:1, attemptNo:-1}

progressLogs

internId	ObjectId, indexed
date	date // user-declared activity da
loggedAt	datetime // server timestamp – do not
enrollmentId	ObjectId, nullable
assignmentId	ObjectId, nullable
hoursSpent	number 0.5-12
completionPct	int, nullable
notes	string
hasBlocker	boolean
blockerText	string, nullable
blockerResolvedAt	datetime, nullable

Index: {internId:1, date:-1}, {hasBlocker:1, blockerResolvedAt:1}

feedback

submissionId	ObjectId, nullable
progressLogId	ObjectId, nullable
internId	ObjectId, indexed
mentorId	ObjectId, indexed
rating	int 1-5
comments	string, min 20 chars

```
actionItems      [string]
decision         enum: approve | request_changes | no_action
acknowledgedAt   datetime, nullable
editableUntil    datetime
parentFeedbackId ObjectId, nullable    // for threads
```

milestones

```
title, description, targetDate
scope          enum: cohort | individual
cohortId / internId
linkedEnrollments [ObjectId]
linkedAssignments [ObjectId]
requiresSignOff   boolean
signedOffBy, signedOffAt
status           enum: on_track | at_risk | missed | complete
```

notifications (*owned by Java service*)

```
userId, type, channel (email|in_app), subject, body
relatedEntityType, relatedEntityId, deepLink
status          enum: queued | sent | failed | skipped | read
attempts        int
lastError       string
scheduledFor, sentAt, readAt
idempotencyKey  string, unique
```

reports (*owned by Java service*)

```
type, parameters (object), format (json|csv|pdf)
requestedBy, requestedAt
status enum: pending | processing | completed | failed
fileUrl, expiresAt
```

riskScores (owned by DS service)

```
internId, indexed
asOfDate      date, indexed
score         number 0-1
band          enum: on-track | watch | at-risk | insuff
features      object    // the raw feature vector used
topFactors    [{ factor, impact, description }]
modelVersion  string
thresholdVersion string
```

Index: {internId:1, asOfDate:-1}

auditLogs

```
actorId, actorRole, action, entityType, entityId
before, after      (objects, PII-minimised)
ipAddress, userAgent, traceId, timestamp
```

Retention: 1 year.

files

```
originalName, storedName, mimeType, sizeBytes
ownerId, relatedEntityType, relatedEntityId
storagePath, uploadedAt
```

8. API Contracts

Each service publishes an OpenAPI 3 spec in `/docs/api/`. The spec is written and reviewed **before** implementation (ARCH-6); QA builds the Postman collection from it and MERN builds against a generated mock.

8.1 Core API (MERN) — /api/v1

POST	/auth/login		POST	/auth/ref
POST	/auth/logout		POST	/auth/for
POST	/auth/reset-password		POST	/auth/cha
GET	/auth/me			
GET	/users	POST	/users	GET /u
PATCH	/users/:id	DELETE	/users/:id (soft)	POST /u
GET	/users/:id/summary			
GET	/cohorts	POST	/cohorts	GET/PATCH /coho
GET	/cohorts/:id/interns			
GET	/courses	POST	/courses	GET/PATCH/DELET
GET	/skill-tags	POST	/skill-tags	
GET	/enrollments?internId=&courseId=&status=&overdue=			
POST	/enrollments	POST	/enrollments/bulk	
PATCH	/enrollments/:id	DELETE	/enrollments/:id	
GET	/assignments?cohortId=&internId=&status=&dueBefore			
POST	/assignments	PATCH	/assignments/:id	
POST	/assignments/:id/publish			
GET	/submissions?assignmentId=&internId=&status=			
POST	/assignments/:id/submissions			
GET	/submissions/:id			
GET	/progress-logs?internId=&from=&to=&hasBlocker=			
POST	/progress-logs	PATCH/DELETE	/progress-log	
GET	/feedback?internId=&mentorId=&unacknowledged=			
POST	/feedback	POST	/feedback/:id/acknowl	
GET	/milestones?cohortId=&internId=			
POST	/milestones	PATCH	/milestones/:id	
POST	/milestones/:id/sign-off			

GET	/dashboard/intern	GET	/dashboard/mentor
GET	/dashboard/admin?cohortId=		
GET	/mentor/review-queue		
POST	/files	GET	/files/:id
GET	/audit-logs	GET/PATCH	/settings
GET	/health	GET	/health/ready

Service-to-service read endpoints (role SERVICE, used by Java and DS — designed for bulk, not per-record chattiness):

```
GET /internal/v1/cohorts/:id/snapshot?from=&to=
    → full denormalised cohort state: interns, enrollmen
      submissions, progressLogs, feedback. Paginated. Th
      endpoint DS uses for feature building.

GET /internal/v1/interns/:id/activity?from=&to=
GET /internal/v1/overdue?cohortId=&asOf=
GET /internal/v1/mentors/:id/interns
```

8.2 Notification & Reporting Service (Java) — /api/v1

```
GET /notifications?userId=&status=&unreadOnly=
POST /notifications/:id/read
POST /notifications/read-all
GET /notifications/unread-count

GET /notification-preferences/:userId
PUT /notification-preferences/:userId

POST /jobs/:jobName/trigger          (admin/service only –
GET /jobs/status                     (last run, next run, o

POST /reports                        (returns reportId; asy
GET /reports/:id
GET /reports/:id/download
```

GET /reports?type=&requestedBy=

GET /health | GET /health/ready

8.3 Analytics Service (DS) — /api/v1

GET /scoring/interns/:id → current score,
POST /scoring/interns/:id/refresh
GET /scoring/interns/:id/history?from=&to=
GET /scoring/cohorts/:id → all interns' b
POST /scoring/batch → trigger nightl
GET /model/metadata

GET /analytics/cohorts/:id/trends?from=&to=
GET /analytics/cohorts/:id/time-to-completion
GET /analytics/cohorts/:id/skill-gaps
GET /analytics/cohorts/:id/engagement-heatmap
GET /analytics/cohorts/:id/feedback-distribution
GET /analytics/cohorts/:id/bottlenecks
GET /analytics/cohorts/compare?ids=a,b (stretch)

GET /health | GET /health/ready

8.4 Integration contracts (INT)

ID	Contract	Owner → Consumer	Due
INT-01	/internal/v1/ cohorts/:id/snapshot schema agreed and mocked	MERN → DS, JAVA	End of Sprint 1
INT-02	Service-token auth scheme implemented and documented	MERN → JAVA, DS	Sprint 1
INT-03	Notification list + unread-count contract	JAVA → MERN	Start of Sprint 2

INT-04	Risk score response shape (score, band, topFactors[], modelVersion, asOfDate)	DS → MERN, JAVA	Start of Sprint 2
INT-05	Chart-data JSON shapes for each insight panel	DS → MERN	Start of Sprint 3
INT-06	Report request/response + async job polling contract	JAVA → MERN	Start of Sprint 3
INT-07	Shared error envelope and x- Request-Id propagation	ALL	Sprint 1

Contract-change rule: any change to a published contract requires a note in the daily standup and an update to the OpenAPI spec in the same PR. Silent contract drift is the single most likely way this project loses a week.

9. Non-Functional Requirements

Performance

ID	Requirement
NFR-P1	Dashboard endpoints respond in < 800 ms p95 with seeded data (30 interns, 6 weeks).
NFR-P2	List endpoints < 500 ms p95 at 25 records per page.
NFR-P3	First contentful paint < 2.5 s on a standard office connection.
NFR-P4	The system supports 50 concurrent users without degradation — sized for an intern programme, not a public product.

NFR-P5	Nightly batch scoring for 200 interns completes in < 5 minutes.
NFR-P6	Weekly digest generation for 50 mentors completes in < 3 minutes.
NFR-P7	No N+1 query patterns on list endpoints; verified by review before Sprint 3.

Security

ID	Requirement
NFR-S1	HTTPS everywhere in shared environments; HTTP redirects to HTTPS.
NFR-S2	OWASP Top 10 addressed: injection, broken access control, XSS, SSRF, insecure deserialisation, security misconfiguration.
NFR-S3	All rich-text input sanitised server-side (allow-list HTML) before storage and before rendering.
NFR-S4	Rate limiting: 5 login attempts / 15 min per account; 100 requests / min per authenticated user; 20 / min per IP on unauthenticated endpoints.
NFR-S5	Secrets in environment variables only. No credentials, tokens, or connection strings in the repository — enforced by a pre-commit secret scan.
NFR-S6	Security headers: CSP, HSTS, X-Content-Type-Options, X-Frame-Options, Referrer-Policy.
NFR-S7	CORS restricted to known frontend origins.
NFR-S8	Uploaded files served through an authorisation-checking endpoint, never from a public static path (see FR-ASG-09).
NFR-S9	Dependency vulnerability scan in CI; no known high/critical vulnerabilities at release.

NFR-S10	Access-control test cases are mandatory for every role-sensitive endpoint (see QA-05). Broken access control is the most likely real defect class in a role-based app built by a small team.
---------	--

Data protection and privacy

ID	Requirement
NFR-D1	This application holds performance-adjacent data about identifiable junior employees. Access is strictly limited to the intern, their mentor, and L&D. There is no "all interns" browse for mentors.
NFR-D2	Risk scores and analytics are not exportable in a form that ranks named interns against one another outside the L&D view.
NFR-D3	Aggregate views suppress groups smaller than 5 (FR-INS-09).
NFR-D4	Interns can view all data held about them (self-service export of their own record).
NFR-D5	Passwords, tokens, and full request bodies are never written to application logs.
NFR-D6	Test and demo environments use synthetic data only, never real intern records copied from production.

Reliability and operations

ID	Requirement
NFR-R1	Target 99% availability during working hours across the programme.
NFR-R2	Daily automated database backup with a documented, <i>tested</i> restore procedure. An untested backup is not a backup.

NFR-R3	Scheduled jobs are idempotent and safely re-runnable (FR-NTF-10).
NFR-R4	Graceful degradation per ARCH-7.
NFR-R5	Each service starts cleanly from <code>docker-compose</code> up with no manual steps beyond a <code>.env</code> file.

Usability and accessibility

ID	Requirement
NFR-U1	WCAG 2.1 AA for colour contrast, keyboard navigation, focus indicators, form labels, and alt text.
NFR-U2	Responsive down to 768 px (tablet). Full mobile layout is a stretch goal.
NFR-U3	Browser support: current and previous major versions of Chrome, Edge, Firefox, Safari.
NFR-U4	Any destructive action requires confirmation and states exactly what will be affected.
NFR-U5	Error messages tell the user what to do next, not just what failed.
NFR-U6	Risk bands use colour plus an icon and text label — never colour alone (accessibility, and it prevents a red dot being read as a verdict).

Maintainability

ID	Requirement
NFR-M1	Linting and formatting enforced in CI per language (ESLint/Prettier, Checkstyle/Spotless, Ruff/Black).
NFR-M2	README per service: purpose, setup, env vars, run, test, troubleshoot.

NFR-M3	Architecture Decision Records for significant choices, in <code>/docs/adr/</code> .
NFR-M4	No hardcoded URLs, thresholds, or magic numbers — configuration only.
NFR-M5	Unit test coverage $\geq 70\%$ on business-logic layers (services, scoring, notification rules). Coverage is a floor, not a goal.

10. QA Requirements and Test Strategy — QA Owner

10.1 Strategy

Test pyramid: many unit tests (owned by each developer), a strong API layer (owned by QA), and a thin UI automation layer covering critical journeys only. QA owns strategy, integration/API testing, UI automation, regression, and defect management on PearlTrack; developers own unit tests for their own code. QA does not become the person who writes everyone's tests — that pattern fails by Sprint 2.

10.2 Requirements

ID	Pri	Requirement
QA-01	M	Test plan document: scope, approach, environments, entry/exit criteria, risks, schedule. Reviewed by the PO in Sprint 1.
QA-02	M	Test cases in PearlTrack, traceable to requirement IDs from this document. Every M requirement has at least one positive and one negative case.

QA-03	M	Requirements traceability matrix: requirement ID → test case IDs → status. This is the demo-day evidence that scope was met.
QA-04	M	Postman collection covering all Core API endpoints: happy path, validation failures, auth failures, boundary values. Environment files for local and shared.
QA-05	M	Role-permission test suite: for every role-sensitive endpoint, attempt access as each of the four roles and assert the expected 200/403. This matrix is the highest-value test asset in the project.
QA-06	M	Newman runs the Postman collection in CI on every PR to main.
QA-07	M	UI automation with Playwright [ASSUMPTION – chosen over Selenium for setup speed, built-in waiting, and trace viewer; switch only if the team has existing Selenium assets]. Page Object Model, no hardcoded sleeps.
QA-08	M	Automated critical journeys: intern login → view assignment → submit; mentor login → review queue → give feedback; admin login → create user → enroll in course; intern logs progress → mentor sees it.
QA-09	M	Smoke suite runs in under 10 minutes and gates every deploy to the shared environment.
QA-10	S3	Full regression suite executed before each sprint demo, with a signed-off report.
QA-11	M	Defect workflow on PearlTrack: New → Triaged → In Progress → Ready for Test → Verified → Closed, with Reopened and Won't Fix. Every bug needs steps to reproduce, expected vs actual, environment, severity, and evidence.

QA-12	M	Severity definitions: S1 blocker (core flow unusable, data loss, auth bypass) — fix immediately; S2 major (feature broken, no workaround) — fix in sprint; S3 minor (workaround exists) — backlog; S4 cosmetic. Severity is QA's call; priority is the PO's.
QA-13	M	Exit criteria per sprint: zero open S1, zero open S2 on demoed features, all M acceptance criteria verified, regression pass rate $\geq 95\%$, traceability matrix updated.
QA-14	S3	Integration test suite covering MERN $\leftrightarrow$ Java and MERN $\leftrightarrow$ DS, including failure modes: service down, slow response, malformed payload, expired service token. Testing the unhappy integration path matters more than the happy one — ARCH-7 is only real if it has been tested with the dependency actually switched off.
QA-15	S3	Notification testing: trigger jobs manually (FR-NTF-11), verify content against a checklist, verify idempotency by re-running, verify preferences suppress correctly, capture emails via Mailhog.
QA-16	S3	Analytics validation: independently recompute a sample of metrics (e.g. in a spreadsheet) and compare against the API output. Charts that render beautifully with wrong numbers are worse than no charts.
QA-17	S3	Basic performance check with k6 or JMeter against NFR-P1/P2 at 50 concurrent users.
QA-18	S3	Security test checklist: the access-control matrix, IDOR attempts on <code>/:id</code> endpoints, XSS payloads in every rich-text field, file-upload type and size bypass, rate-limit verification, direct file-URL access.
QA-19	M	Accessibility check with axe DevTools on every major screen; no critical violations at release.
QA-20	M	Cross-browser check of critical journeys on Chrome, Firefox, and Safari before demo day.

QA-21	M	Test data management: reset-and-seed script (FR-SYS-07) runs before each regression cycle so results are reproducible.
QA-22	S3	Demo-day dry run: full walkthrough on the demo environment with the demo dataset, 48 hours before, with a rollback plan.
QA-23	M	Bug bash: 90 minutes with the whole team at the end of each sprint, before the demo.

10.3 Environments

Env	Purpose	Data	Owner
Local	Development	Seeded synthetic	Each developer
Shared/QA	Integration and QA	Seeded synthetic, reset weekly	QA
Demo	Sprint demos and demo day	Curated demo dataset, frozen 48h before	QA + Scrum Master

11. Sprint Plan

Sprint 0 — Setup (first 3–5 days, overlapping Sprint 1)

Owner	Work
ALL	Repo, branching, CI skeleton, docker-compose, <code>.env.example</code> , PearlTrack board and workflow, Definition of Ready/Done agreed

ALL	Contract workshop: agree entity names, the shared error envelope, the auth scheme, and INT-01/INT-02/INT-07. Half a day here saves a week in Sprint 3.
MERN	Project scaffold, MongoDB connection, base middleware
JAVA	Spring Boot scaffold, scheduler proof-of-concept, Mailhog wired up
DS	Python service scaffold, synthetic data generator started (FR-SCR-10)
QA	Test plan draft, PearlTrack defect workflow, Postman workspace

Sprint 1 (Weeks 1–2) — Foundation

Owner	Deliverables
MERN	FR-AUTH-01→09, 11, 12 · FR-USR-01→04, 06→08 · FR-CRS-01→07 · FR-SYS-01→03, 07
JAVA	Service scaffold, health endpoints, service-token auth, scheduler running a no-op job, email send proven end-to-end with a template
DS	FR-SCR-10 complete · FR-SCR-01 feature pipeline against synthetic data · exploratory analysis notebook
QA	QA-01, QA-02 (auth + user + course cases), QA-04 partial, QA-05 matrix defined, QA-11 workflow live
Sprint goal	A user can log in with the right role and see a course catalog and their assigned courses.

Sprint 2 (Weeks 3–4) — Core loop

Owner	Deliverables
-------	--------------

MERN	FR-ASG-01→09, 11 · FR-PRG-01→05, 07 · FR-FBK-01→05, 08 · FR-MLS-01→03 · FR-DSH-01→04, 10→12, 20→22, 25 · FR-SYS-08 · <code>/internal/v1/*</code> endpoints
JAVA	FR-NTF-04, 05, 07, 09, 11 · digest logic built against the snapshot endpoint
DS	FR-SCR-02 rule-based scorer live behind <code>/scoring/interns/:id</code> · FR-SCR-04, 05, 11 · DS-2 begins FR-INS-01, 02
QA	QA-04 complete, QA-05 executed, QA-07/08 automation for the three critical journeys, QA-09 smoke suite in CI
Sprint goal	The full loop works: intern logs progress and submits → mentor reviews and gives feedback → L&D sees it on the overview. This is the MVP. If nothing else ships, this must.

Sprint 3 (Weeks 5–6) — Intelligence and integration

Owner	Deliverables
MERN	FR-DSH-05, 13, 23, 24 · FR-NTF-07 UI · FR-ADM-02→05 · FR-SYS-09 · integration of all Java and DS panels
JAVA	FR-NTF-01, 02, 03, 06, 08, 10, 12 · FR-RPT-01→04, 06→09
DS	FR-SCR-06→08, 12, 13 · FR-INS-03→09
QA	QA-10 full regression, QA-14 integration failure modes, QA-15, QA-16, QA-17, QA-18, QA-19, QA-20, QA-22
Sprint goal	Mentors receive real weekly digests, overdue items are flagged automatically, and L&D sees live cohort analytics.

Weeks 7–8 (if the programme runs to 8) — Stretch and hardening

Priority order if time is short: **(1) hardening and bug fixes, (2) PDF export, (3) at-risk ML model, (4) AI summaries.** Ship a solid tool over a feature-rich broken one. The stretch items are deliberately last because each is a nice demo moment, and none of them matters if the core loop is flaky.

Also in this window: demo-day preparation, README and handover docs, retrospective, and an adoption plan for the next intern batch.

12. Dependencies and Critical Path

#	Dependency	Risk if late	Mitigation
1	MERN auth + data model → everything	Total blockage; three people idle	MERN delivers the OpenAPI spec and a mock server in Sprint 0, before implementation. Java and DS build against the mock.
2	Real activity data → DS model	No data to train or demo on	FR-SCR-10 synthetic generator in Sprint 1. Do not wait for real data.
3	/internal/v1/cohorts/:id/snapshot → Java digests + DS features	Two services blocked at once	This is the highest-priority MERN endpoint after auth. Target: end of Sprint 1.
4	DS risk score contract → MERN dashboard panels	Rework in Sprint 3	INT-04 frozen at the start of Sprint 2, mocked immediately.
5	Skill tag vocabulary → skill-gap analysis	FR-INS-03 becomes unbuildable	Tag list defined with the PO in Sprint 1.

6	SMTP access → all email notifications	Digests cannot be demoed	Mailhog locally from Sprint 0; request real relay access in week 1, not week 5.
7	Stable MERN UI → Playwright automation	Constant test breakage	Stable <code>data-testid</code> attributes on every interactive element — a MERN requirement, not a QA request. Agree in Sprint 0.

Single biggest structural risk: one MERN developer is the dependency for four other people. Two mitigations are non-optional — mock-first contracts (ARCH-6), and the Java/DS developers taking frontend panels for their own features in Sprint 3 rather than queuing behind MERN.

13. Definition of Ready / Definition of Done

Ready (a story can enter a sprint): requirement ID referenced · acceptance criteria written and testable · dependencies identified and available or mocked · API contract agreed if cross-service · design/ wireframe attached if UI · estimated by the team.

Done (a story can be demoed): acceptance criteria met · unit tests written and passing · code reviewed and merged to `main` · API documented in the OpenAPI spec · QA test case written and executed · no open S1/S2 defects · deployed to the shared environment · loading/ empty/error states implemented · PearlTrack ticket updated.

14. Risks

Risk	Impact	Likelihood	Mitigation
MERN becomes the bottleneck	High	High	Mock-first contracts; Java/DS own their own UI panels in Sprint 3
Insufficient real data for DS	High	Certain	Synthetic data generator in Sprint 1 (FR-SCR-10)
Scope creep from stretch goals	Medium	High	Stretch items are explicitly last; PO enforces the MVP-first order
Integration left to Sprint 3	High	Medium	Contracts in Sprint 0; a thin end-to-end vertical slice (login → API → dummy Java call) demoed in Sprint 1
Interns learning the stack while building	Medium	High	Sprint 1 scope is deliberately light; pair with mentors; timebox spikes
QA saturated across four domains	Medium	High	Developers own unit tests; QA focuses on API, integration, and critical-path UI
Rotating Scrum Master causes process drift	Low	Medium	Written handover checklist between sprints; PO is constant
Email relay unavailable	Medium	Medium	Mailhog for dev and demo; request access in week 1
Risk scores misread as judgements about people	Medium	Medium	FR-SCR-05 explainability, FR-SCR-13 use-policy note, NFR-U6 labelling, and PO framing at demo day

Demo-day environment failure	High	Low	Frozen demo dataset, dry run 48h ahead, recorded backup walkthrough
------------------------------	------	-----	---

15. Demo Day and Handover Checklist

- ☐ End-to-end scripted walkthrough covering all three roles
 - ☐ Demo dataset frozen and verified 48 hours ahead (QA-22)
 - ☐ Recorded backup video in case of live failure
 - ☐ Each domain presents its own piece, including the integration point it depends on
 - ☐ QA presents the traceability matrix and defect summary — including what is knowingly *not* done
 - ☐ DS presents model logic and limitations honestly, not just the accuracy number
 - ☐ READMEs, OpenAPI specs, and ADRs complete in `/docs`
 - ☐ Known issues and backlog handed to L&D for the next batch
 - ☐ Retrospective: what to keep, drop, and try — captured in PearlTrack
-

16. Open Questions for the Product Owner

Answer these before Sprint 1 planning. Each has a proposed default so work is not blocked.

#	Question	Proposed default
1	Should interns see their own risk band?	Yes — band and supportive guidance only, no numeric score (FR-AUTH-14)

2	Is the insights dashboard embedded in the MERN app, or a separate Streamlit/Dash app?	Embedded (FR-INS-08). Decide in Sprint 1, not Sprint 3
3	Is there an SMTP relay available, or is Mailhog the ceiling?	Request relay in week 1; Mailhog as fallback
4	Is SSO available and in scope?	Out of scope unless the IdP is available immediately
5	One mentor per intern, or many-to-many?	One primary, optional secondary (FR-USR-04/05)
6	Are real intern names and data used, or fully synthetic through demo day?	Synthetic until the tool is formally adopted (NFR-D6)
7	Who hosts the shared and demo environments, and who has deploy rights?	Needs an answer in Sprint 0
8	Is Playwright acceptable, or is Selenium required for training alignment?	Playwright (QA-07)
9	Will the tool be formally adopted for the next batch, and who owns it afterwards?	Affects how much hardening is worth doing in weeks 7–8
10	Is the AI summary feature permitted, and with which model/endpoint?	Stretch only; confirm data-handling approval before any implementation
11	Should mentor turnaround metrics be visible to mentors themselves, or L&D only?	Visible to the mentor for their own data; L&D sees all (FR-DSH-22)
12	Retention: how long is cohort data kept after the programme ends?	1 year, then archive and anonymise

Appendix A — Requirement Count by Priority

Priority	Count (approx.)	Sprint
M (MVP)	78	1–2
S3	62	3
ST (Stretch)	21	7–8 or dropped
NFR	38	Continuous
QA	23	Continuous

Appendix B — Suggested PearlTrack Epic Structure

EPIC-01	Authentication & Access Control	(MERN)
EPIC-02	User, Cohort & Mentor Management	(MERN)
EPIC-03	Course Catalog & Enrollment	(MERN)
EPIC-04	Assignments & Submissions	(MERN)
EPIC-05	Progress Logging	(MERN)
EPIC-06	Mentor Review & Feedback	(MERN)
EPIC-07	Dashboards	(MERN + DS)
EPIC-08	Notification Service	(JAVA)
EPIC-09	Reporting Service	(JAVA)
EPIC-10	Progress Scoring Model	(DS-1)
EPIC-11	Insights Dashboard	(DS-2)
EPIC-12	Platform, Admin & Audit	(MERN)
EPIC-13	Quality & Test Automation	(QA)
EPIC-14	DevOps, Environments & Release	(ALL)
EPIC-15	Stretch: AI Summaries, PDF, ML Model	(ALL)